

DEVS Model and Simulation of Single and Combined Attacks on an Industrial Wastewater System

Braeden Kloke

SYSC 5104: Methodologies for Discrete Event Modeling and Simulation

Department of Systems and Computer Engineering

Carleton University

Dr. Gabriel Wainer

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1. Conceptual model

Industrial wastewater systems need to control the amount of residual chlorine in their discharged wastewater to remain in compliance with government regulations and prevent harm to the environment. The PLCs of these systems can be compromised by spoofing and tampering attacks and cause the system to discharge wastewater that exceeds safety limits. Taking inspiration from [1], we can study the single and combined effects of these attacks and evaluate their impact on the system's ability to remain in compliance. Table 1 describes the components for the system.

Table 1: Component descriptions.

Component	Description
Wastewater	Wastewater flowing out of the system.
Chemical Analyzer	Measures total residual chlorine in the wastewater.
PLC	Sends commands to the chemical feed pump based on wastewater samples.
Chemical Feed Pump	A valve that opens or closes; when open, it releases sodium bisulphite into the wastewater.
Spoof Attacker	Spoofs messages to the PLC.
Tamper Attacker	Tampers incoming messages to the PLC.
Vulnerable PLC	Coupled model of attackers and PLC.

2. DEVS model

This section will describe the DEVS models and tests for the system. For brevity, diagrams and test descriptions for the components Wastewater, Chemical Analyzer, and Chemical Feed Pump have been omitted.

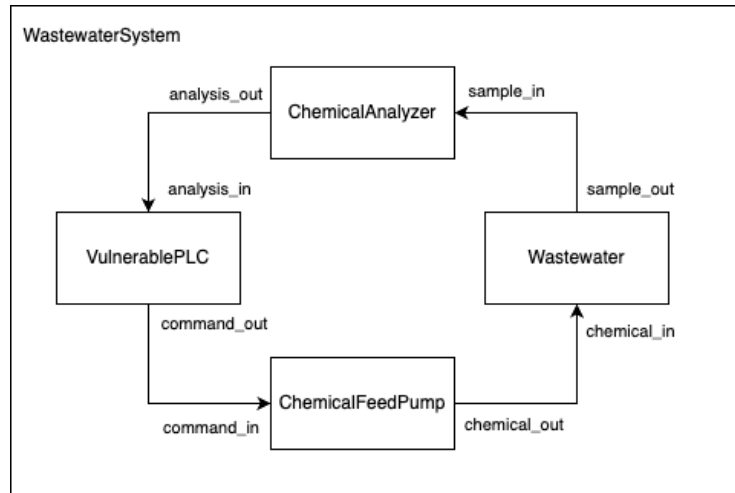


Figure 1: Coupled DEVS model of a wastewater system. The system controls the pH level of the wastewater such that the pH level remains within an upper and lower bound. Wastewater flows from a wastewater treatment plant and periodically increases its pH level due to the plant's sanitization process. Wastewater samples are analyzed by the Chemical Analyzer which transmits results to the Vulnerable PLC. The Vulnerable PLC sends commands to the Chemical Feed Pump which dispenses chemicals into the Wastewater to lower its pH level.

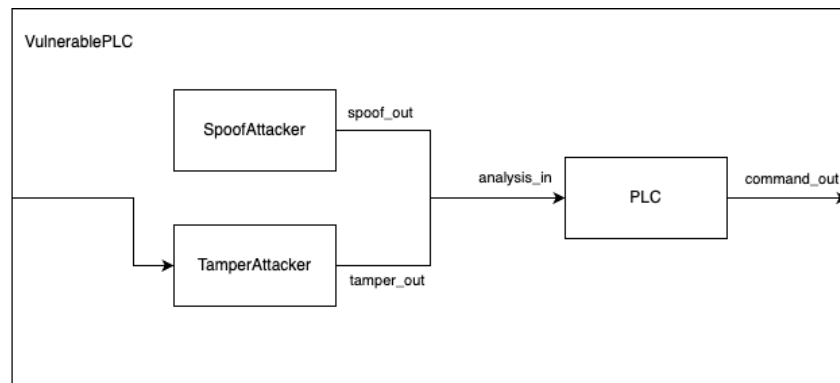


Figure 2: Coupled DEVS model of a Vulnerable PLC. The objective of the attackers is to make the PLC send false commands. Messages to the PLC are intercepted by the Tamper Attacker before being passed onto the PLC. The Spoof Attacker periodically spoofs messages to the PLC.

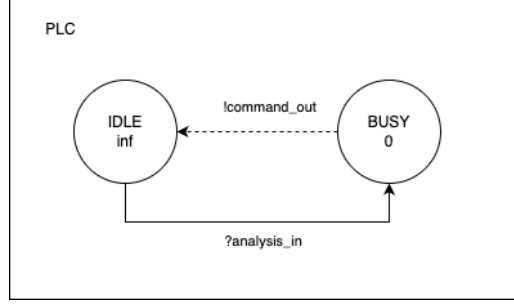


Figure 3: DEVS graph of a PLC. Calculates and outputs commands based on input analysis.

Table 2: Atomic DEVS model of a PLC. Calculates commands based on the pH safety limit and a tolerance. If the input analysis is too close to the pH safety limit, then it outputs an ON command to the Chemical Feed Pump.

DEVS Element	Definition
$X =$	$\{ analysis_in \in \mathbb{R}^+ \}$
$Y =$	$\{ command_out \in \{ON, OFF\} \}$
$S =$	$\{ phase \in \{IDLE, BUSY\}, analysis \in \mathbb{R}^+, pH_safety_limit \in \mathbb{R}^+, tolerance \in \mathbb{R}^+ \}$
$ta(S) =$	if $phase = BUSY \rightarrow 0$; else $\rightarrow \infty$;
$\delta_{ext}(S, e, X) =$	if $phase = IDLE \rightarrow$ $analysis = analysis_in.value$; $phase = BUSY$; else $doNothing()$;
$\delta_{int}(S) =$	if $phase = BUSY \rightarrow phase = IDLE$; else $doNothing()$;
$\lambda(S) =$	if $analysis > pH_safety_limit - tolerance \rightarrow command_out.value = ON$; else $\rightarrow command_out.value = OFF$;

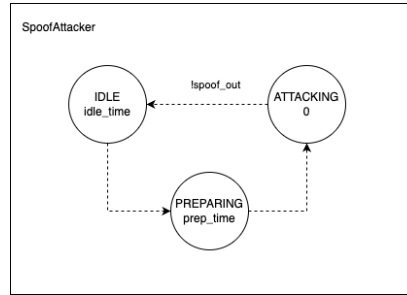


Figure 4: DEVS graph of a Spoof Attacker. Periodically spoofs messages.

Table 3: Atomic DEVS model of a Spoof Attacker.

DEVS Element	Definition
$X =$	$\emptyset$
$Y =$	$\{ spoof_out \in \mathbb{R}^+ \}$
$S =$	$\{ phase \in \{IDLE, PREPARING, ATTACKING\}, prep_time \in \mathbb{R}^+, idle_time \in \mathbb{R}^+ \}$
$ta(S) =$	if $phase = PREPARING \rightarrow prep_time$; else if $phase = IDLE \rightarrow idle_time$; else $\rightarrow 0$;
$\delta_{ext}(S, e, X) =$	$doNothing()$;
$\delta_{int}(S) =$	if $phase = ATTACKING \rightarrow phase = IDLE$; else if $phase = IDLE \rightarrow phase = PREPARING$; else if $phase = PREPARING \rightarrow phase = ATTACKING$;
$\lambda(S) =$	if $phase = ATTACKING \rightarrow spoof_out.value = spoof()$;

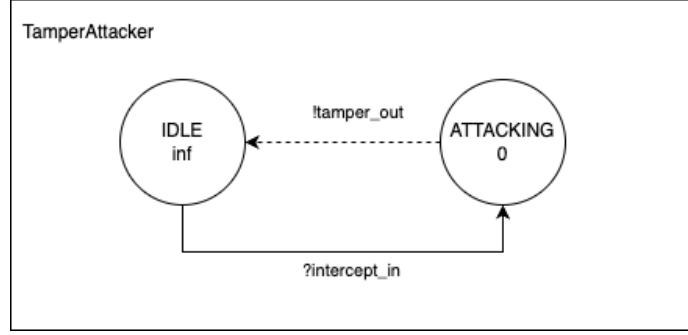


Figure 5: DEVS graph of a Tamper Attacker. Intercepts messages as input and outputs tampered messages.

Table 4: Atomic DEVS model of Tamper Attacker.

DEVS Element	Definition
$X =$	$\{intercept_in \in \mathbb{R}^+\}$
$Y =$	$\{tamper_out \in \mathbb{R}^+\}$
$S =$	$\{phase \in \{IDLE, ATTACKING\}, intercepted_msg \in \mathbb{R}^+, bias \in \mathbb{R}^+\}$
$ta(S) =$	$if\ phase = IDLE \rightarrow \infty;$ $else \rightarrow 0;$
$\delta_{ext}(S, e, X) =$	$intercepted_msg = x;$ $if\ phase = IDLE \rightarrow phase = ATTACKING;$
$\delta_{int}(S) =$	$if\ phase = ATTACKING \rightarrow phase = IDLE;$ $else \rightarrow doNothing();$
$\lambda(S) =$	$if\ phase = ATTACKING \rightarrow tamper_out.value = intercepted_msg - bias;$ $else \rightarrow doNothing();$

Table 5: Atomic DEVS model of a Chemical Feed Pump. When the pump is ON it outputs chemicals every time unit.

DEVS Element	Definition
$X =$	$\{command_in \in \{ON, OFF\}\}$
$Y =$	$\{chemical_out \in \mathbb{R}^+\}$
$S =$	$\{phase \in \{ON, OFF\}, chemical_feed_rate \in \mathbb{R}^+\}$
$ta(S) =$	$if\ phase.ON \rightarrow 1;$ $else \rightarrow \infty;$
$\delta_{ext}(S, e, X) =$	$phase = command_in.value;$
$\delta_{int}(S) =$	$doNothing();$
$\lambda(S) =$	$if\ phase.ON \rightarrow chemical_out.value = chemical_feed_rate;$

Table 6: Atomic DEVS model of Wastewater which periodically increases its pH every time unit. Chemicals can be added to the wastewater to lower its pH level.

DEVS Element	Definition
$X =$	$\{chemical_in \in \mathbb{R}^+\}$
$Y =$	$\{sample_out \in \mathbb{R}^+\}$
$S =$	$\{pH_level \in \mathbb{R}^+, pH_increase_rate \in \mathbb{R}^+\}$
$ta(S) =$	$for\ all\ s \in S \rightarrow 1;$
$\delta_{ext}(S, e, X) =$	$pH_level = pH_level - chemical_in.value;$
$\delta_{int}(S) =$	$pH_level = pH_level + pH_increase_rate;$
$\lambda(S) =$	$sample_out.value = pH_level;$

Table 7: Atomic DEVS model of a Chemical Analyzer. The analyzer samples of wastewater and analyzes the pH level of the sample.

DEVS Element	Definition
$X =$	$\{sample_in \in \mathbb{R}^+\}$
$Y =$	$\{analysis_out \in \mathbb{R}^+\}$
$S =$	$\{phase = \{IDLE, ANALYZING\}, sample \in \mathbb{R}^+, analysis_time \in \mathbb{R}^+\}$
$ta(S) =$	$if\ phase = ANALYZING \rightarrow analysis_time;$ $else \rightarrow \infty;$
$\delta_{ext}(S, e, X) =$	$if\ phase = IDLE \rightarrow sample = sample_in.value;$ $else \rightarrow doNothing();$
$\delta_{int}(S) =$	$if\ phase = ANALYZING \rightarrow phase = IDLE;$ $else \rightarrow doNothing();$
$\lambda(S) =$	$if\ phase = ANALYZING \rightarrow analysis_out.value = analyze(sample);$

3. Model verification

This section describes the tests to verify the model implementations.

```
root@main:~/cadmium-projects/model-wastewater-system# ./bin/plc_test
time,model_id,model_name,port_name,data
0,1,Test File,,0
0,2,PLC,,
0,1,Test File,out,8.9
0,1,Test File,,1
0,2,PLC,,
0,2,PLC,commandOut,0
0,2,PLC,,
1,1,Test File,out,9.1
1,1,Test File,,inf
1,2,PLC,,
1,2,PLC,commandOut,1
1,2,PLC,,
1,1,Test File,,inf
1,2,PLC,,
```

Figure 6: PLC unit test results. For a safety limit of 10.0 and tolerance of 1.0, the decision boundary is calculated to be 9.0. The PLC correctly sends OFF commands (0) for inputs less than the decision boundary and ON commands (1) for inputs greater than the decision boundary.

```
root@main:~/cadmium-projects/model-wastewater-system# ./bin/tamper_attacker_test
time,model_id,model_name,port_name,data
0,1,Test File,,0
0,2,Tamper Attacker,,
0,1,Test File,out,10
0,1,Test File,,inf
0,2,Tamper Attacker,,
0,2,Tamper Attacker,tamperOut,9
0,2,Tamper Attacker,,
0,1,Test File,,inf
0,2,Tamper Attacker,,
```

Figure 7: Tamper attacker unit test results. The tamper attacker correctly subtracts a bias from intercepted messages.

```

root@main:~/cadmium-projects/model-wastewater-system# ./bin/wastewater_system_no_attacks
time,model_id,model_name,port_name,data
0,2,Tamper Attacker,,
0,3,Spoof Attacker,,
0,4,PLC,,
0,5,ChemicalAnalyzer,,
0,6,Wastewater,,
0,7,ChemicalFeedPump,,
1,5,ChemicalAnalyzer,,
1,6,Wastewater,sampleOut,9
1,6,Wastewater,,
2,5,ChemicalAnalyzer,,
2,6,Wastewater,sampleOut,9.1
2,6,Wastewater,,
3,2,Tamper Attacker,,
3,5,ChemicalAnalyzer,analysisOut,9
3,5,ChemicalAnalyzer,,
3,6,Wastewater,sampleOut,9.2
3,6,Wastewater,,
3,2,Tamper Attacker,tamperOut,9
3,2,Tamper Attacker,,
3,4,PLC,,
3,4,PLC,commandOut,0
3,4,PLC,,

```

Figure 8: System test results. The system correctly keeps the wastewater pH level below the safety limit.

4. Experiment results

This section describes the experiments performed on the system.

Table 8: Experimental frame.

Invariants	PLC safety limit = 10.0; PLC tolerance = 1.0; Chemical feed rate = 0.2; Wastewater pH increase rate = 0.1
Variables	Tamper bias; Spoof idle time
Assumptions	pH level of wastewater increases linearly

Table 9: Experiment descriptions.

ID	Name	Description
1	No attacks.	System in normal conditions.
2	Spoof attack.	System under a spoofing attack.
3	Tamper attack.	System under a tampering attack.
4	Combined attacks.	System under both spoofing and tampering attacks.

```

root@main:~/cadmium-projects/model-wastewater-system# ./bin/wastewater_system_no_attacks
time,model_id,model_name,port_name,data
0,2,Tamper Attacker,,
0,3,Spoof Attacker,,
0,4,PLC,,
0,5,ChemicalAnalyzer,,
0,6,Wastewater,,
0,7,ChemicalFeedPump,,
1,5,ChemicalAnalyzer,,
1,6,Wastewater,sampleOut,9
1,6,Wastewater,,
2,5,ChemicalAnalyzer,,
2,6,Wastewater,sampleOut,9.1
2,6,Wastewater,,
3,2,Tamper Attacker,,
3,5,ChemicalAnalyzer,analysisOut,9
3,5,ChemicalAnalyzer,,
3,6,Wastewater,sampleOut,9.2
3,6,Wastewater,,
3,2,Tamper Attacker,tamperOut,9
3,2,Tamper Attacker,,
3,4,PLC,,
3,4,PLC,commandOut,0
3,4,PLC,,

```

Figure 9: Partial experiment 1 results. System behaves as expected under normal conditions and maintains the pH level of the wastewater below safety limits for 100 time units.

```

root@main:~/cadmium-projects/model-wastewater-system# ./bin/wastewater_system_spoof_attack
time,model_id,model_name,port_name,data
0,2,Tamper Attacker,,
0,3,Spoof Attacker,,
0,4,PLC,,
0,5,ChemicalAnalyzer,,
0,6,Wastewater,,
0,7,ChemicalFeedPump,,
1,3,Spoof Attacker,,
1,5,ChemicalAnalyzer,,
1,6,Wastewater,sampleOut,9
1,6,Wastewater,,
2,3,Spoof Attacker,,
2,5,ChemicalAnalyzer,,
2,6,Wastewater,sampleOut,9.1
2,6,Wastewater,,
2,3,Spoof Attacker,spoofOut,8
2,3,Spoof Attacker,,
2,4,PLC,,
2,4,PLC,commandOut,0
2,4,PLC,,
2,7,ChemicalFeedPump,,
3,2,Tamper Attacker,,
3,3,Spoof Attacker,,
3,5,ChemicalAnalyzer,analysisOut,9
3,5,ChemicalAnalyzer,,
3,6,Wastewater,sampleOut,9.2
3,6,Wastewater,,
3,2,Tamper Attacker,tamperOut,9
3,2,Tamper Attacker,,
3,4,PLC,,
3,4,PLC,commandOut,0
3,4,PLC,,
3,7,ChemicalFeedPump,,
4,3,Spoof Attacker,,
4,5,ChemicalAnalyzer,,
4,6,Wastewater,sampleOut,9.3
4,6,Wastewater,,
4,3,Spoof Attacker,spoofOut,8

```

Figure 10: Partial experiment 2 results. Spoof attacker with an idle time of 1.0 has no impact on the system after 100 time units.


```

26,6,Wastewater,sampleOut,9.9
26,6,Wastewater,,
27,2,Tamper Attacker,,
27,5,ChemicalAnalyzer,analysisOut,9.8
27,5,ChemicalAnalyzer,,
27,6,Wastewater,sampleOut,10
27,6,Wastewater,,
27,2,Tamper Attacker,tamperOut,9.2
27,2,Tamper Attacker,,
27,4,PLC,,
27,4,PLC,commandOut,1
27,4,PLC,,
27,7,ChemicalFeedPump,,
28,5,ChemicalAnalyzer,,
28,6,Wastewater,sampleOut,10.1
28,6,Wastewater,,
28,7,ChemicalFeedPump,chemicalOut,0.2
28,7,ChemicalFeedPump,,
29,2,Tamper Attacker,,
29,5,ChemicalAnalyzer,analysisOut,10
29,5,ChemicalAnalyzer,,
29,6,Wastewater,sampleOut,10
29,6,Wastewater,,
29,7,ChemicalFeedPump,chemicalOut,0.2
29,7,ChemicalFeedPump,,
29,2,Tamper Attacker,tamperOut,9.4
29,2,Tamper Attacker,,

```

Figure 11: Partial experiment 3 results. Tamper attacker with a bias of 0.6 has some impact on the system after 100 time units with the pH level oscillating between safe and unsafe levels.

```

97,5,ChemicalAnalyzer,analysisOut,10
97,5,ChemicalAnalyzer,,
97,6,Wastewater,sampleOut,10
97,6,Wastewater,,
97,2,Tamper Attacker,tamperOut,9.4
97,2,Tamper Attacker,,
97,4,PLC,,
97,4,PLC,commandOut,1
97,4,PLC,,
97,7,ChemicalFeedPump,,
98,3,Spoof Attacker,,
98,5,ChemicalAnalyzer,,
98,6,Wastewater,sampleOut,10.1
98,6,Wastewater,,
98,7,ChemicalFeedPump,chemicalOut,0.2
98,7,ChemicalFeedPump,,
98,3,Spoof Attacker,spoofOut,8
98,3,Spoof Attacker,,
98,4,PLC,,
98,4,PLC,commandOut,0
98,4,PLC,,
98,7,ChemicalFeedPump,,
99,2,Tamper Attacker,,
99,3,Spoof Attacker,,
99,5,ChemicalAnalyzer,analysisOut,10
99,5,ChemicalAnalyzer,,
99,6,Wastewater,sampleOut,10
99,6,Wastewater,,
99,2,Tamper Attacker,tamperOut,9.4
99,2,Tamper Attacker,,

```

Figure 12: Partial experiment 4 results. Under combined attacks with a spoofer idle time of 1.0 (Experiment 2) and a tamper bias of 0.6 (Experiment 3) the system consistently remains at unsafe pH levels.

References

- [1] A. Jawad and J. Jaskolka, "Single and Combined Cyberattack Impact on Industrial Wastewater Systems," in *2023 10th International Conference on Dependable Systems and Their Applications (DSA)*, Tokyo, Japan, 2023.